

ANS PHARMACOLOGY

MBS230-P&T

UNIT 2.5

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2.5-MUSCARINIC RECEPTOR ANTAGONISTS

MUSCARINIC RECEPTOR ANTAGONISTS

LS - 2.5: LEARNING OBJECTIVES:

1. Classify Anti Muscarinic Drugs
2. Describe mechanism of action, pharmacological effects, clinical uses, unwanted effects and contraindications of muscarinic receptor antagonists
3. Describe treatment of Atropine toxicity

MUSCARINIC RECEPTOR ANTAGONISTS

Introduction

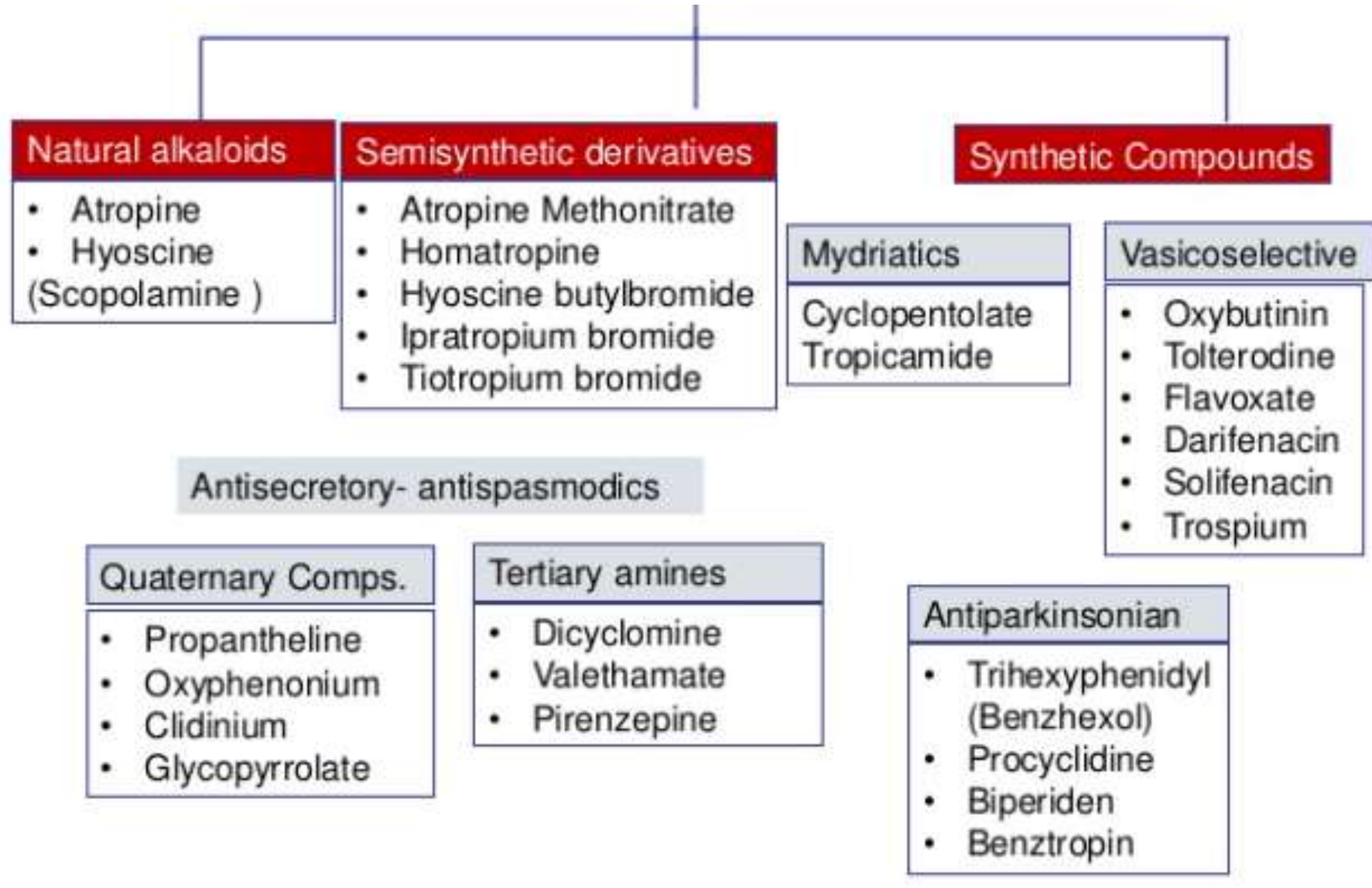
- The **classical muscarinic antagonists** are derived from plants.
- The deadly nightshade (*Atropa belladonna*), a relative of the tomato and potato, contains atropine. Jimson weed (*Datura stramonium*) is even more widespread in Tennessee than the deadly nightshade.
- The tiny dark seeds from the Jimson weed pod are sometimes ingested for their hallucinogenic effect, a central side effect of atropine-like substances.
- Henbane (*Hyoscyamus niger*) contains primarily scopolamine and hyoscine.

MUSCARINIC RECEPTOR ANTAGONISTS

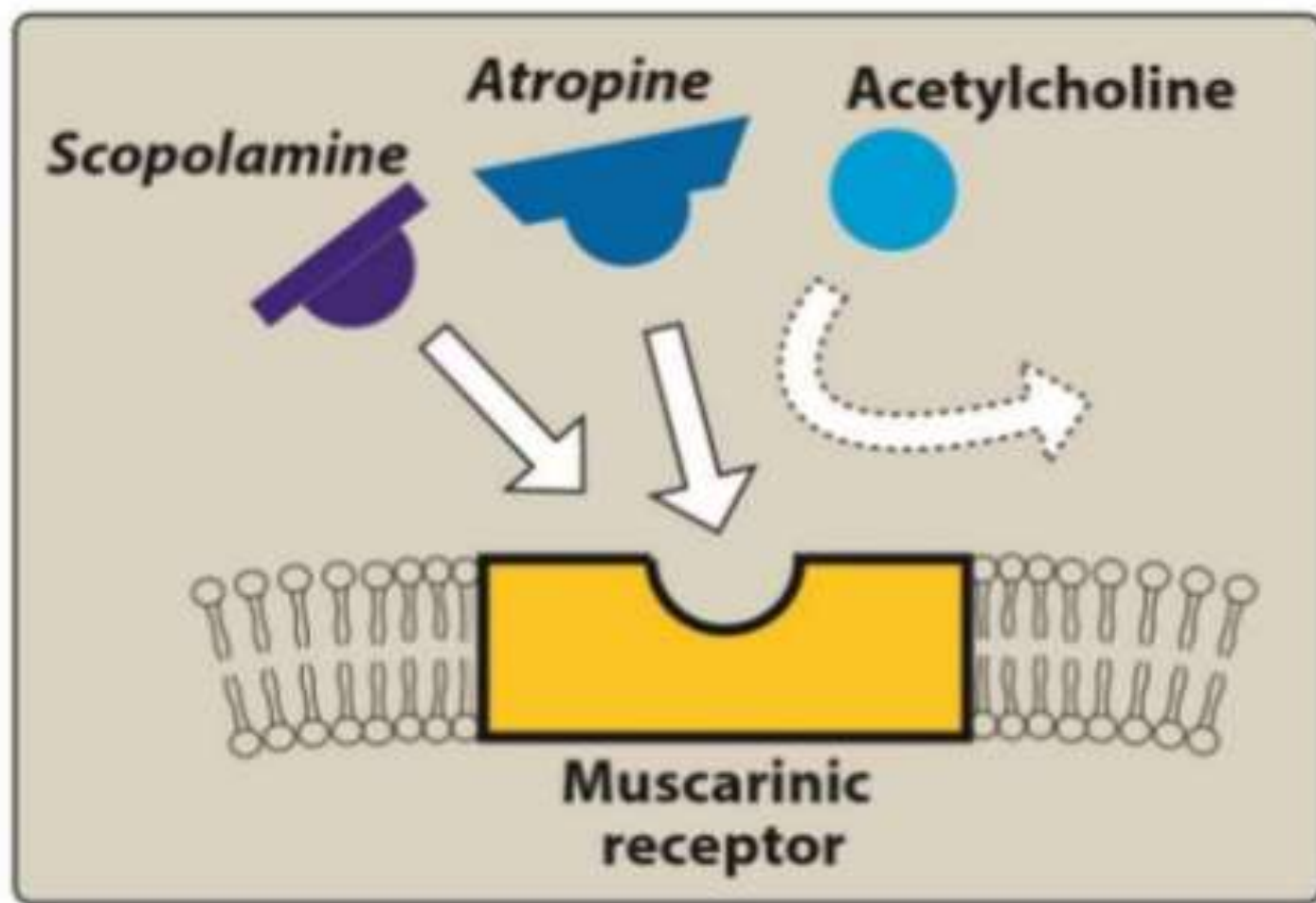
Muscarinic receptor antagonists (parasympatholytic) are competitive antagonists of Ach at all muscarinic receptors (m1 through m5)

- Natural occurring compounds with anti Muscarinic effects have been known and used for millennia as medicines, poisons and cosmetics. The prototype is atropine
- Different tissues respond differently to muscarinic antagonists: salivary, sweat and bronchial glands are the most sensitive and can be blocked by low doses of atropine. Parietal cells in the stomach are the most resistant and the block of gastric acid secretion requires high doses of atropine.

Classification of Anti Muscarinic Drugs



Mechanism of action of anticholinergic drugs



Pharmacological Actions of Atropine

Central Nervous System

- Mild CNS stimulant and longer lasting sedation action in therapeutic doses
- Large doses:- excitement, hallucinations, restlessness and coma
- Antiparkinsonian effect by reducing Ach overactivity in basal ganglia (Reduce tremors of Parkinsonism disease)
- Anti motion sickness effect by suppressing vestibular disturbances (e.g. Scopolamine).

Pharmacological Actions of Atropine

- Low doses of Atropine causes bradycardia due blockade of m1 receptors on vagal nerve endings (inhibit release of Ach)
- Moderate to high doses cause tachycardia due to blockade of m2 receptors in the heart
- ↑SA and AV nodal conduction and PR interval (due to blockade of muscarinic receptors in ↓AV node)

Pharmacological Actions of Atropine

Glands

- All secretions under cholinergic influence are reduced (due to M₃ receptor blockade)
- Except milk and bile secretion
- Skin and mucus membrane become dry

Pharmacological Actions of Atropine

Smooth Muscles

GIT

- Tone and Motility
- Sphincter tone may cause constipation
- Relaxes smooth muscle of gall bladder

GUT

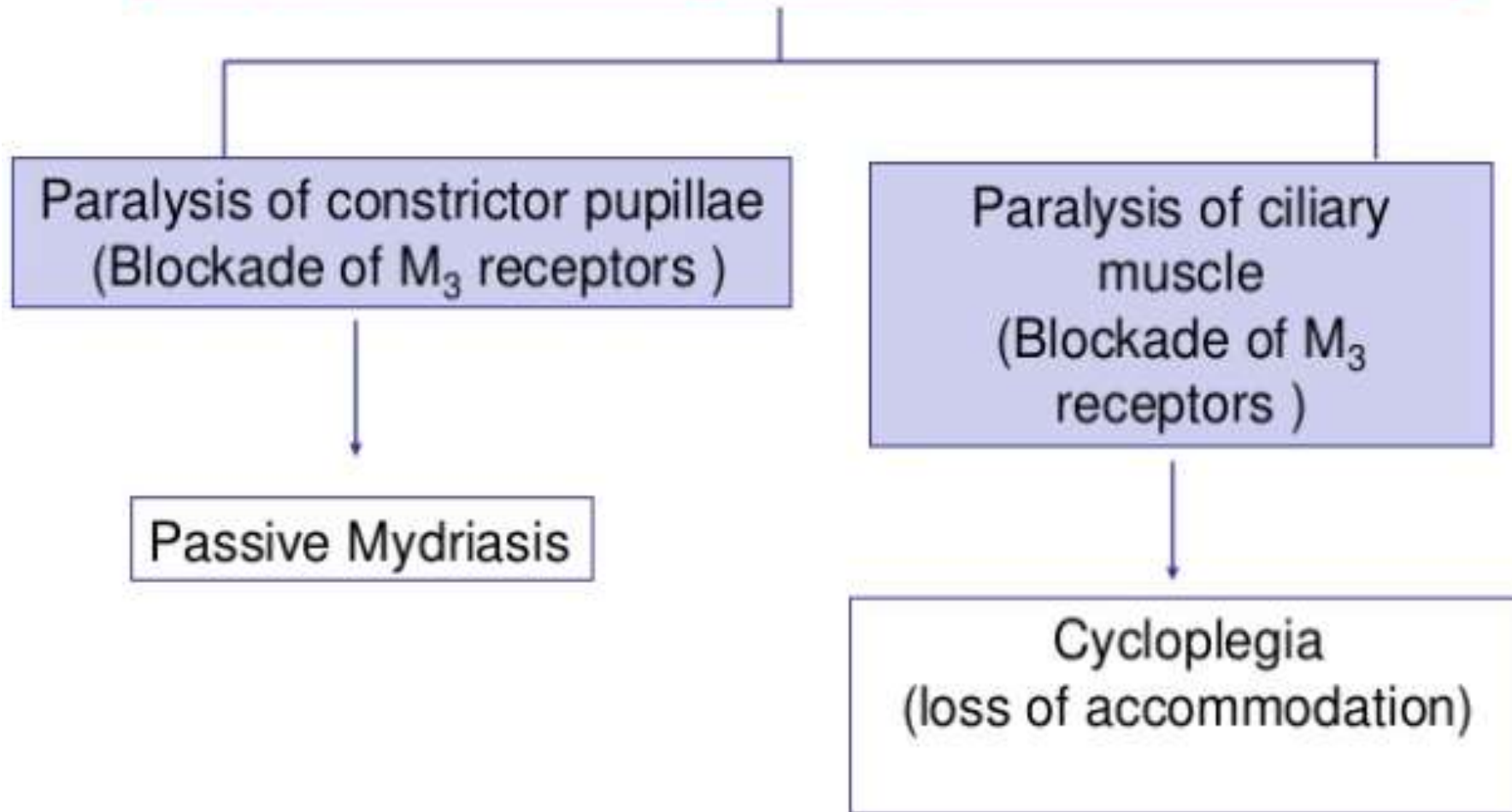
- Relaxes detrusor muscles of bladder & tone of trigone and sphincter (Retention of urine)

Bronchi

- Relaxes bronchial smooth muscle
- Secretion and muco-ciliary clearance (mucus plug may form)

Pharmacological actions of atropine

Atropine action on eye on topical administration



- It also causes reduction in lacrimal secretion leading to patients complaining of dry or 'sandy' eyes

Pharmacokinetics of anticholinergic drugs

- Absorption :
 - Atropine, scopolamine other tertiary amines well absorbed
- Distribution:
 - except quaternary compounds rest drugs are widely distributed.
- Metabolism:
 - 50% atropine & 80 % scopolamine metabolized by liver as conjugates
- Excretion
 - 50% atropine excreted unchanged in urine
 - $t_{1/2} = 3$ hours

MUSCARINIC RECEPTOR ANTAGONISTS: PHARMACOLOGICAL EFFECTS

Effector	Effects
Heart	Increased heart rate
Lacrimation	Decreases
Respiratory system	Bronchodilatation Decreased tracheobronchial secretions
GIT	Reduced peristaltic activity and motility, resulting in prolonged gastric emptying and intestinal transit Decreased salivation (dry mouth) Decreased gastric secretions Relaxation of biliary tract smooth muscles
Urinary tract	Relaxation of the ureter and bladder smooth muscle Sphincter constriction
Sweat glands	Decreased sweat sweating (dry skin)
Eye	Paralysis of the ciliary muscle and loss of accommodation (cycloplegia) Pupillary dilatation (mydriasis) Increased intraocular pressure (may precipitate glaucoma in some patients)
CNS	In high doses, atropine stimulates the CNS resulting in restlessness, headache, excitement, hallucinations and delirium. Hyoscine produces sedation and drowsiness.

CLINICAL USES OF MUSCARINIC RECEPTOR ANTAGONISTS

- For eye examination (testing errors of refraction and fundoscopic examination): homatropine, cyclopentolate and tropicamide (given topically)
- Anterior uveitis and iritis - used alternatively with miotics to break adhesions between the iris and the lens: topical homatropine
- To provide rest to the iris in iritis, iridocyclitis, keratitis and after partial iridectomy: topical homatropine, cyclopentolate and tropicamide

CLINICAL USES OF MUSCARINIC RECEPTOR ANTAGONISTS CONT'D

- To reduce urinary urgency and relieve bladder spasm in inflammatory bladder disorders: hyoscine butyryl bromide, dicyclomine, oxybutinin, tolterodine, and trospium
- Bradycardia: atropine (parenteral)
- Prevention of motion sickness: hyoscine (a.k.a scopolamine)
- Bronchial asthma: ipratropium and tiotropium (by inhalation)

CLINICAL USES OF MUSCARINIC RECEPTOR ANTAGONISTS CONT'D

- Peptic ulcer disease: pirenzepine
- To suppress bronchiolar secretions during general and spinal anaesthesia: glycopyrrolate, atropine and hyoscine
- Parkinson's disease and parkinsonism: benztropine, trihexyphenidyl and benzhexol (centrally acting anti-muscarinics)
- Muscarinic excess (e.g. in organophosphate poisoning): atropine

MUSCARINIC RECEPTOR ANTAGONISTS CONT'D

Adverse effects

The adverse effects are extensions of pharmacological activity: mydriasis, cycloplegia, blurred vision, dry eyes, tachycardia, dry mouth, elevated body temperature, dry skin, urinary retention, agitation, hallucinations and delirium

Contraindications

Glaucoma, GI and urinary obstruction (e.g. prostate hypertrophy)

Treatment of muscarinic receptor antagonist overdose

Physostigmine is used

Atropine Toxicity

- Hot as a hare; hyperpyrexia
- Red as beetroot; cutaneous vasodilation
- Dry as a bone; dry skin
- Blind as a bat; mydriasis and cycloplegia
- Hallucinations, restlessness and excitement

Treatment of Atropine Poisoning

- Hospitalization
- Gastric lavage
- Tepid sponging
- Diazepam to control convulsions
- Antidote for atropine- physostigmine 1-4mg injected slowly

Contraindications of Atropine

- Acute Congestive Glaucoma
- Elderly patients (prone to urinary retention)

Classification of anticholinergics according to clinical use

S.N	Use	Drug
1	Ophthalmic use	Cyclopentolate, tropicamide, homatropine
2	Antispasmodic	Dicyclomine, Hyoscine methyl bromide, propantheline
3	Urinary antispasmodic	Oxybutinin, flavoxate, darifenacin, trospium
4	Bronchodilation	Ipratropium, tiotropium
5	Parkinsonism	benztropine, biperidine, benzhexol

Summary

- Introduction
- Classification
- Pharmacological actions
- Therapeutic uses
- Adverse reactions
- Contraindications

Short Answer Questions

What is the Rational of;

- Use of atropine in preanesthetic medication
- Preference of physostigmine over neostigmine in treatment of OPP (office of pesticides Program)
- Scopolamine in motion sickness

What are the differences between;

- Atropine and Scopolamine
- Ocular actions of Atropine and Ephedrine

Name;

- Specific Blockers of m1 receptors
- Specific blockers of m3 receptors

Why

- Atropine is used in children for mydriasis?
- Tropicamide/Cyclopentolate is preferred in adult patients of mydriasis?

LEARNING OUTCOMES

Students must be able to comprehend:

Classification of antimuscarinic drugs, mechanism of action, Pharmacological effects, clinical uses, unwanted effects and contraindications of muscarinic receptor antagonists

END

Thanks for listening